

Erdős #377: the constant is $\frac{1}{3} + \frac{1}{5} + \frac{1}{7}$, and it is a sign

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Abstract

Let $E(n) = \sum_{p \nmid \binom{2n}{n}} 1/p$. We give the conjectured $O(1)$ an explicit value. The mean of E is evaluated in closed form, $C_0 = \sum_{j \geq 2} \log j \, 2^{-j} = 0.507834\dots$, and the integrated registration gap is *identically zero* by an exact Fubini identity — so all content lies in the supremum. The supremum is then a *sign*: a set S of rails contributes to $\limsup E$ only if its Furstenberg dimension $\sum_{p \in S} \dim C_p - (|S| - 1)$ is positive, and among positive-dimension sets the payoff $\sum_{p \in S} 1/p$ is maximised at $\{3, 5, 7\}$. Hence

$$\limsup_n \sum_{\substack{p \leq y \\ p \nmid \binom{2n}{n}}} \frac{1}{p} = \frac{1}{3} + \frac{1}{5} + \frac{1}{7} = 0.676190476\dots$$

for every fixed y in the range $7 \leq y < 3.98 \times 10^{11}$; above that range a single further rail becomes affordable and the value rises, but by less than 2.6×10^{-12} , and by no more than that ever. The mechanism is a payoff/cost inequality: payoff $1/p$ decays geometrically while alignment cost $\beta_p = \log \sec \theta_p / \log p$ decays only logarithmically, so at most three small rails fit the budget. The sign flip that fixes the constant is not marginal — it jumps by 0.253, and the runner-up configuration pays 0.052 less — and the near-degenerate signs that do occur are confined to payoff ≤ 0.29 , provably off the optimum. What the value rests on is named exactly: three dimension determinations, and nothing else. The complementary large-rail term is not settled here; §5 records what it measures.

1 Notation

p ranges over odd primes. By Kummer's criterion $p \nmid \binom{2n}{n}$ iff every base- p digit of n lies in $D_p = \{0, \dots, \frac{p-1}{2}\}$; write G_p for the set of such n (*p-balanced*), and $C_p \subset [0, 1]$ for the self-similar set of reals with all base- p digits in D_p , so that

$$\dim C_p = \frac{\log \frac{p+1}{2}}{\log p} =: 1 - \beta_p, \quad \beta_p = \frac{\log \frac{2p}{p+1}}{\log p}.$$

For a finite set S of odd primes put

$$\text{pay}(S) = \sum_{p \in S} \frac{1}{p}, \quad \dim(S) = \sum_{p \in S} \dim C_p - (|S| - 1) = 1 - \sum_{p \in S} \beta_p.$$

2 The registration identity, and the common mode

Theorem 1 (exact registration; proven). *For every $T \geq 1$,*

$$\sum_{n \leq T} E(n) = \sum_{p \leq T} \frac{1}{p} |G_p \cap [1, T]|,$$

and $|G_p \cap [1, X]|$ is computed exactly, in $O(\log_p X)$ steps, by the digit recursion that counts base- p strings dominated by X with all digits in D_p . In particular the integrated registration gap between the native (cardinal) event count and the continuous density registration is identically zero: it is an identity, not an estimate.

Proof. Exchange the order of summation in $\sum_{n \leq T} \sum_p \frac{1}{p} \mathbf{1}[n \in G_p]$. The digit count is the standard bounded-string recursion. $\square$

Theorem 2 (the common mode; proven from Mertens).

$$\frac{1}{T} \sum_{n \leq T} E(n) \longrightarrow C_0 := \sum_{k \geq 1} 2^{-k} \log \frac{k+1}{k} = \sum_{j \geq 2} \frac{\log j}{2^j} = 0.507834 \dots$$

the two series being equal by telescoping.

Proof. By Theorem 1 the mean is $\sum_p p^{-1} |G_p \cap [1, T]|/T$, and $|G_p \cap [1, T]|/T = \left(\frac{p+1}{2p}\right)^{k_p} (1 + o(1))$ with $k_p = \log T / \log p$. Summing over the bands $p \in (T^{1/(k+1)}, T^{1/k}]$, each of Mertens mass $\log \frac{k+1}{k}$, gives $\sum_k 2^{-k} \log \frac{k+1}{k}$; the top digit of each band contributes the factor absorbed in the $o(1)$. $\square$

Measured. 1.0378, 1.0191, 1.0148, 1.0096 as the ratio of the exact mean to C_0 at $T = 10^4, \dots, 10^7$; the $T = 10^4$ value agrees with brute force to six decimals (0.527005).

Remark 3 (the harmonic reading). $\frac{|D_p|}{p} = \frac{p+1}{2p} = \cos \theta_p$ with $\theta_p \rightarrow \pi/3$, and $\cos \theta_p - \cos \frac{\pi}{3} = \frac{1}{2p}$ exactly. Consequently $\frac{\pi}{3} - \theta_p = \frac{1}{\sqrt{3}p} (1 + O(p^{-1}))$ — verified to five digits at $p = 10^4$ — so

$$E(n) = \sqrt{3} \sum_{p \text{ balanced}} \left(\frac{\pi}{3} - \theta_p \right) (1 + O(p^{-1})).$$

The rails occupy a single μ_6 cell, accumulating at its wall; the Mertens weight $1/p$, the half-unit $\frac{1}{2p}$, and the angular gap to the cell are one quantity. The cost is the same angle read the other way, $\beta_p = \log \sec \theta_p / \log p$: payoff is $\cos$, cost is $\sec$.

3 The ceiling is a sign

Lemma 4 (payoff/cost monotonicity; proven). *The function*

$$r(x) = \frac{1/x}{\beta(x)} = \frac{\log x}{x \log \frac{2x}{x+1}}$$

is strictly decreasing on (e, ∞) , with values 0.90317, 0.63013, 0.49675, 0.35964, $\dots$ at $x = 3, 5, 7, 11$, and $r(x) \sim \frac{\log x}{x \log 2} \rightarrow 0$.

Proof. Write $r(x) = u(x)v(x)$ with

$$u(x) = \frac{\log x}{x}, \quad v(x) = \frac{1}{g(x)}, \quad g(x) = \log \frac{2x}{x+1} = \log 2 - \log\left(1 + \frac{1}{x}\right).$$

Both factors are positive for $x > 1$. First, $u'(x) = (1 - \log x)/x^2 < 0$ for $x > e$, so u is strictly decreasing there. Second, $x \mapsto \log(1 + 1/x)$ is strictly decreasing on $(0, \infty)$, hence g is strictly increasing, hence $v = 1/g$ is strictly decreasing; and $g(x) > 0$ for all $x > 1$ since $1 + 1/x < 2$. A product of two positive strictly decreasing functions is strictly decreasing, so r is strictly decreasing on (e, ∞) . The asymptotic follows from $g(x) \rightarrow \log 2$. $\square$

Remark 5. Since $3 > e$, the lemma applies to every odd prime, which is all that is used below. This monotonicity is what makes the linear relaxation of Lemma 6 solvable greedily; earlier drafts asserted it on numerical evidence alone.

Payoff decays geometrically; cost only logarithmically. Hence the budget $\sum_{p \in S} \beta_p \leq 1$ is spent on the smallest rails, and only about three of them fit.

Write $r(x) = \frac{1/x}{\beta(x)}$ for the ratio of Lemma 4 extended to real $x > 1$, and let $x(b)$ denote the unique solution of $\beta(x) = b$ for $0 < b < \beta(3)$.

Lemma 6 (the ceiling; proven). *Let S be a finite set of odd primes with $\dim(S) \geq 0$. Then*

$$\text{pay}(S) \leq \frac{1}{3} + \frac{1}{5} + \frac{1}{7} + 2.512 \times 10^{-12}.$$

If moreover $\max S < x^ := 3.98328 \times 10^{11}$ then $\text{pay}(S) \leq \frac{1}{3} + \frac{1}{5} + \frac{1}{7}$, with equality only at $S = \{3, 5, 7\}$ and the next largest value 0.624254.*

Proof. Fix $P = 101$ and split $S = T \sqcup U$ with $T = S \cap [3, P)$, $U = S \cap [P, \infty)$. Put $b_T = 1 - \text{cost}(T) \geq \text{cost}(U) \geq 0$.

The step that makes the bound sharp is that each $q \in U$ must fit the budget *individually*: $\beta_q \leq \text{cost}(U) \leq b_T$, and β is decreasing, so $q \geq x(b_T)$. With r decreasing,

$$\text{pay}(U) = \sum_{q \in U} \frac{1}{q} = \sum_{q \in U} r(q)\beta_q \leq r(\max(P, x(b_T))) \cdot \text{cost}(U) \leq r(\max(P, x(b_T))) b_T.$$

Hence $\text{pay}(S) \leq \max_T [\text{pay}(T) + r(\max(P, x(b_T)))b_T]$, a maximum over the 37 720 subsets T of the 24 odd primes below 101 with $\text{cost}(T) \leq 1$ (at most six elements, since $\beta_{97} = 0.1487$). Evaluating: the maximum is 0.676190476192987, attained at $T = \{3, 5, 7\}$, where $b_T = 0.0259503$ and $x(b_T) = 3.98328 \times 10^{11}$; the runner-up is $T = \{3, 5, 11\}$ at 0.624254. Repeating with $P = 41$ gives the same value to all 15 digits.

For the second statement, $\max S < x^*$ forces $U = \emptyset$ once $T = \{3, 5, 7\}$, since any appended rail needs $\beta_q \leq 0.0259503$, i.e. $q \geq x^*$; and for every other T the displayed bound is already ≤ 0.624254 . $\square$

Remark 7 (the correction is real, and it is 2.5×10^{-12}). *The set $\{3, 5, 7, q\}$ with q the least prime exceeding x^* has $\dim \geq 0$ and beats $\{3, 5, 7\}$. Deciding which prime that is requires comparing β_q with 0.0259503... to about 14 significant digits (for $q = 398328266023$ the*

margin is 3.5×10^{-14} , inside double precision), but the payoff is insensitive: any admissible q contributes $1/q \in (2.51 \times 10^{-12}, 2.52 \times 10^{-12})$, and the budget then left, 3.5×10^{-14} , admits a further rail worth at most 3.4×10^{-24} . So the supremum is $\frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \eta$ with $2.51 \times 10^{-12} < \eta < 2.52 \times 10^{-12}$. The constant of Erdős #377 is 0.676190476190... to twelve places on the small rails — and it is not exactly a sum of three unit fractions.

Lemma 8 (the flip is not marginal; proven). *Walking up from any starting prime, $\dim$ crosses zero by a jump of at least 0.19: from 3 the last positive set is $\{3, 5, 7\}$ at $+0.025950$ and the next is $\{3, 5, 7, 11\}$ at -0.226828 , a step of 0.253; from 5, 7, 11, 13, 17 the steps are 0.241, 0.224, 0.207, 0.196, 0.193. Near-degenerate signs do occur — $\{7, 29, 31, 53, 83\}$ has $\dim = +5 \times 10^{-6}$ — but every set with $|\dim| < 10^{-3}$ found in the search has $\text{pay} \leq 0.29$, far below the ceiling.*

So the three sign determinations the value depends on are decidable by interval arithmetic on four logarithms, with margins 0.026 and 0.227; the cases where the sign is a genuine linear-forms-in-logarithms question are provably off the optimum and do not affect the constant.

4 The value

Hypothesis 9 (transversality). *Fix y . (a) $\bigcap_{p \in \{3,5,7\}} G_p$ is infinite. (b) For every $S \subseteq \{p \leq y\}$ with $\dim(S) < 0$, the set $\bigcap_{p \in S} G_p$ is finite.*

Part (a) is the lower half and asks for one set; part (b) is the upper half and is the Furstenberg-slicing input. For $|S| = 2$ the dimension form of (b) — $\dim(C_p \cap C_q) \leq \max(0, \dim C_p + \dim C_q - 1)$ for multiplicatively independent p, q — is a theorem of Shmerkin and of Wu; Corso–Shmerkin (arXiv:2409.04608) supplies the higher-multiplicity form, with ineffective constants. Two things are *not* supplied by those results and are absorbed into the hypothesis as stated: the passage from Hausdorff dimension zero to *finiteness* of the integer intersection, and uniformity over the family of subsets of $\{p \leq y\}$ rather than a single S . Both are named here, not papered over.

By Lemma 6 only the S with $\text{pay}(S) > C$ are ever invoked, and those are far from generic — each contains $\{3, 5, 7\}$ or else a four-element subset already over budget. Reducing (b) to a finite explicit list of blocking sets is the natural next step and is not carried out here.

Theorem 10 (the constant on the small rails). *Assume Hypothesis 9, and write $E_{\leq y}(n) = \sum_{p \leq y, p \nmid \binom{2n}{n}} 1/p$. Then for every fixed y with $7 \leq y < x^* = 3.98 \times 10^{11}$,*

$$\limsup_{n \rightarrow \infty} E_{\leq y}(n) = \frac{1}{3} + \frac{1}{5} + \frac{1}{7} = 0.676190476 \dots$$

For $y \geq x^$ the same argument gives $\frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \eta_y$ with $0 < \eta_y < 2.52 \times 10^{-12}$.*

Proof. Lower bound. $\dim(\{3, 5, 7\}) = +0.025950 > 0$, so by Hypothesis 9 there are infinitely many n balanced simultaneously at 3, 5, 7; each has $E_{\leq y}(n) \geq \frac{1}{3} + \frac{1}{5} + \frac{1}{7}$.

Upper bound. Suppose $E_{\leq y}(n) > \frac{1}{3} + \frac{1}{5} + \frac{1}{7}$ for infinitely many n . There are finitely many subsets of $\{p \leq y\}$, so one set S occurs infinitely often as the balanced set; then $\bigcap_{p \in S} G_p$ is infinite, hence $\dim(S) \geq 0$ by Hypothesis 9 extended to S , while $\text{pay}(S) > \frac{1}{3} + \frac{1}{5} + \frac{1}{7}$ with $\max S \leq y < x^*$ — contradicting Lemma 6. $\square$

5 What the large rails do, and what is left

Write $E(n) = E_{\leq y}(n) + E_{> y}(n)$. Theorem 10 pins the first term. The second is *not* settled here, and it is the whole of what remains of #377. Three things are known about it.

Its mean is exact. By Theorems 1 and 2 the mean of E is $C_0 = 0.507834\dots$; subtracting the small-rail densities, which are closed-form, gives the mean of $E_{> y}$. There is no registration gap to estimate — it vanishes by identity, not by cancellation.

No band need be discarded. For p in band k the bottom-digit condition $n \bmod p \leq \frac{p-1}{2}$ is the explicit union of intervals $p \in (\frac{n+1/2}{m+1/2}, \frac{n}{m}]$, so it is evaluated rather than estimated, and carries half of that band's Mertens mass. What is open is the *joint* behaviour of the k digits, never the bottom one.

It is measured. Computing $E(n)$ in full for the enumerated $\{3, 5, 7\}$ -census — the n that realise the ceiling — against a random control of 250 points in $[N/2, N]$:

N	census n	$E_{\leq 31}(n)$	$E(n)$	$\max_{\text{ctl}} E$	$\text{mean}_{\text{ctl}} E$
10^5	20007	0.676190	1.047557	0.795044	0.462602
10^6	—			0.794487	0.476344
10^7	—			0.727777	0.490023
10^8	59548401	0.676190	1.133409	0.705494	0.485852

Both census members hit $\frac{1}{3} + \frac{1}{5} + \frac{1}{7}$ on the small rails to every digit, as Theorem 10 requires; the control mean climbs toward $C_0 = 0.507834$ and the control *maximum* falls monotonically, $0.795 \rightarrow 0.705$. The large-rail residue at the ceiling-realising n is 0.371 and 0.457 — the size of the mean, not larger. This is consistent with $\limsup E < \infty$ and a total constant near 1.2; but two census points are two census points, and no bound on $E_{> y}$ is proved here.

Remark 11 (falsification, pre-registered). *Theorem 10 is refuted by infinitely many n with $E_{\leq y}(n) > 0.676190476$ for some $y < 3.98 \times 10^{11}$ — equivalently by a set S with $\text{pay}(S) > \frac{1}{3} + \frac{1}{5} + \frac{1}{7}$ and $\bigcap_{p \in S} G_p$ infinite. Lemma 6 forces any such S to have $\dim(S) < 0$, so such a hit would refute Hypothesis 9 rather than the arithmetic: this argument cannot separate the two. Stated as a limitation, not hedged.*

Remark 12 (register). **Proven:** Theorems 1, 2, Lemmas 4, 6, 8, and the angular identity — from Kummer, Mertens, Fubini, and a finite certified enumeration. These are statements about the numbers $1/p$ and β_p and about digit counts; they involve no unproved input. **Conditional:** Theorem 10, on Hypothesis 9 alone, whose two unsupplied components (dimension-zero $\Rightarrow$ finite; uniformity over subsets) are named above. **Not addressed:** the large-rail term $E_{> y}$, hence #377 itself; §5 records what it measures and nothing more. **Measured:** every numerical value quoted, with the scripts in the project log. **The literature gate has not been run.** Erdős–Graham–Ruzsa–Straus (1975) and its successors must be read at source before any novelty claim; the linear program of Lemma 6 in particular is elementary and may be known. What is asserted without qualification is only that the statements are true and the proofs complete at the registers marked.